Product Requirements Document (PRD)

وئام | Wiaam

Unified Patient Profile & Care Navigation Platform for Saudi Arabia

Saudi-first digital health platform • Version 1.4 • Updated with Saudi hospital directory seed data, secure sign up / login, MFA for all roles, Arabic/English language switching, appointment booking, and synced calendar workflows

Product Name	وئام | Wiaam
Product Type	National-scale healthcare interoperability and care-navigation platform
Primary Market	Kingdom of Saudi Arabia (KSA)
Primary Users	Patients, Medical Staff, Admin
Core Value Proposition	One patient profile across providers, better care coordination, faster provider discovery, and medication fulfillment visibility
Document Purpose	Define product scope, requirements, workflows, architecture, risks, roadmap, and success metrics

1. Executive Summary

Wiaam (وئام) is a Saudi-first digital health platform that gives every patient in the Kingdom a single longitudinal profile accessible, under strict role-based permissions, across participating hospitals and clinics. Instead of each hospital maintaining a disconnected copy of the same patient record, the platform uses National ID or Iqama as the primary identity anchor and links medical history, medications, insurance, dependents, demographics, and care activity into one canonical profile.

The platform also functions as a care-navigation layer. Patients can discover nearby hospitals, compare ratings, find doctors matched to their symptoms, verify pharmacy medication availability, and get answers to common questions through an AI chatbot or agent. Medical staff can access clinically relevant patient information, review medications and prior history, and coordinate care. Admin users can manage providers, access controls, integrations, quality metrics, and compliance operations.

2. Problem Statement

- Patient data is fragmented across hospitals, clinics, pharmacies, and insurers.
- The same patient may have multiple profiles across institutions, creating duplication, missing history, medication risk, and poor continuity of care.
- Patients struggle to find the right provider, understand which hospital to visit, and verify medication availability before traveling.
- Healthcare staff waste time collecting information that should already exist and may make decisions with incomplete medical context.
- Administrative teams lack a unified governance layer for identity, data quality, permissions, and ecosystem-wide reporting.

3. Product Vision

Create the trusted digital front door for patient-centered healthcare in Saudi Arabia: one identity, one profile, connected care, intelligent navigation, and secure access for every authorized stakeholder.

4. Saudi Identity and Brand Positioning

Wiaam, meaning harmony and alignment, should be positioned as a distinctly Saudi health platform built around trusted digital identity, secure interoperability, and continuity of care across the Kingdom. Saudi identity in the product should not be limited to language translation; it should be reflected in onboarding through National ID and Iqama, Arabic-first information architecture, local healthcare workflows, culturally appropriate UX, and institutional trust signals.

- Primary identity methods shall support National ID and Iqama from day one.
- Arabic shall be the default interface language, with English fully supported.
- Branding, terminology, date and time formats, and address structures should follow Saudi usage norms.
- The product should be designed for national trust, healthcare-grade reliability, and public-private ecosystem interoperability.
  - The interface shall provide a clear Arabic / English language switch on sign-up, login, chatbot, booking, and in-app navigation surfaces across patient, staff, and admin experiences.

5. Product Goals

- Establish a single patient profile per person using National ID or Iqama as the core identifier.
- Provide authorized hospitals and medical staff with a longitudinal view of patient records.
- Help patients discover the right hospital, doctor, and pharmacy faster.
- Reduce duplicate records, repeat diagnostics, and medication errors.
- Improve provider utilization, care coordination, and patient satisfaction.
- Deliver enterprise-grade security, auditing, and role-based access control.
- Offer a multilingual chatbot or agent trained on common healthcare questions and product workflows.
  - Enable secure role-based sign up, login, and mandatory multi-factor authentication across patients, medical staff, and admin users.
  - Allow patients to book appointments and manage a unified calendar of booked, rescheduled, and cancelled visits.

6. Non-Goals (Phase 1)

- Direct diagnosis by the AI chatbot without clinician review.
- Full replacement of existing hospital EMR or EHR systems.
- Cross-border medical record exchange outside KSA in the initial release.
- Advanced claims adjudication and insurer settlement engine in MVP.
- Consumer social features or public health community feeds.

7. User Roles and Personas

7.1 Patients

- Citizens and residents using National ID or Iqama.
- Need a single health profile, doctor discovery, hospital discovery, medication visibility, appointment support, and plain-language guidance.

7.2 Medical Staff

- Doctors, nurses, pharmacists, technicians, care coordinators.
- Need rapid access to verified patient history, medications, allergies, prior encounters, insurance details, and linked dependents where relevant.

7.3 Admin

- Hospital administrators, network operators, compliance teams, platform super admins.
- Need governance over providers, integrations, access rights, data quality rules, ratings moderation, audit trails, and platform analytics.

8. Core Product Modules

Module	Description
Unified Patient Identity & Profile	Single profile anchored by National ID or Iqama with demographics, age, name, occupation, insurance, dependents, contact details, allergies, medication history, diagnosis history, visit history, and consent status.
Provider Discovery	Nearby hospitals and clinics, filters by specialty, insurance acceptance, rating, distance, and service availability.
Doctor Discovery & Matching	Doctor profiles with specialty, subspecialty, languages, years of experience, gender preference, hospital affiliation, ratings, and symptom-based recommendations.
Pharmacy Availability	Medication lookup and pharmacy stock or dispense availability near the patient, with alternatives and insurance relevance where available.
AI Chatbot / Agent	Answers common questions, navigates the product, triages to the right care path, and escalates to human channels when needed.
Clinical Workspace	Medical staff dashboard for patient search, profile review, encounter summaries, medication reconciliation, and care timeline review.
Admin Console	User management, organization onboarding, role assignment, audit logs, provider management, content moderation, and operational reporting.
Authentication & Identity Access	Role-based sign up and login flows for patients, medical staff, and admins, with Saudi identity proofing, multi-factor authentication, device/session controls, and privileged-access policies.
Appointment Booking & Calendar	Patients can book appointments with hospitals and doctors, receive confirmations and reminders, and view all appointments in a synced calendar experience across channels.

9. Detailed Functional Requirements

9.0 Access, Registration, and Authentication

- System shall require multi-factor authentication for every role: patients, medical staff, and admins.
- Patients shall sign up using National ID or Iqama, mobile verification, consent acceptance, and a secure password or passwordless method with OTP as a second factor.

- Medical staff accounts shall be provisioned by approved healthcare organizations or invite-only onboarding and bound to verified staff identities before activation.
- Admin accounts shall be invitation-only, subject to privileged-access approval, and protected with stronger MFA policy such as authenticator app, hardware key, or equivalent enterprise factor.
- Login flows shall support session timeout, device recognition, suspicious-login detection, lockout thresholds, password reset, and account recovery workflows.
- Authentication events shall be fully audited, including successful login, failed login, MFA challenge, password reset, session revocation, and step-up authentication for sensitive actions.
- Sign-up and login screens shall expose an Arabic / English language switch before authentication.

9.1 Patient Profile

- System shall create one master patient profile per unique National ID or Iqama.
- System shall detect and flag probable duplicate profiles for admin review.
- Profile shall include full name, date of birth, age, gender, nationality, occupation, insurance provider, policy details, dependents, emergency contacts, and contact information.
- Profile shall store medical history including diagnoses, surgeries, allergies, chronic conditions, family history, medications, immunizations, and encounter timeline.
- System shall show data provenance for clinical records, including which hospital, clinic, or provider contributed each item.
- Patients shall be able to view their own profile and request corrections for demographic inaccuracies.
- Medical staff shall see profile data according to role, organization, and consent rules.

9.2 Hospital Discovery

- Patients shall view nearby hospitals and clinics in list and map views.
- Search results shall include distance, rating, specialties, insurance acceptance, operating hours, emergency availability, and wait-time or appointment availability where integrated.
- Patients shall filter by city, district, specialty, insurance, rating, language, gender preference, and open-now status.
- System shall support location permissions and manual location entry.

9.2.1 Saudi Hospital Registry Foundation

- System shall treat the uploaded Saudi hospital directory as the initial seed dataset for the national hospital registry, then normalize, deduplicate, geocode, and enrich it for production use.
- Seed records shall ingest and map the source fields for facility name, location, region, type, ownership, bed count, website, and notes into the platform provider master registry.
- Registry operations shall support canonical Arabic name, optional English transliteration, region and city normalization, geolocation coordinates, facility status, verification timestamp, and source provenance.
- Hospital discovery results shall be powered by this registry and later enriched with ratings, specialties, accepted insurance, emergency capability, and real-time operational signals where available.
- The uploaded directory shall be treated as a trusted starting point for coverage, but not as a complete source of live availability, wait times, or pharmacy inventory without integrations.

9.3 Doctor Discovery

- Each doctor profile shall include name, title, specialty, subspecialty, hospital affiliation, languages, gender, years of experience, rating, next available slot, and accepted insurance.
- Patients shall search by doctor name, specialty, symptom, hospital, insurance, language, and availability.
- System shall recommend doctors based on symptoms using a rules-plus-ML ranking approach, with explicit disclaimer that this is guidance, not diagnosis.

9.3.1 Appointment Booking and Calendar Sync

- Patients shall be able to book appointments with doctors, clinics, and hospitals from provider profiles, search results, and referral paths.
- Booking flows shall display available slots, appointment type, hospital location, clinician, preparation instructions, and confirmation status.
- Patients shall be able to reschedule, cancel, or join a waitlist subject to provider policy.
- All booked, rescheduled, and cancelled appointments shall sync to a unified in-app calendar tied to the user account and visible across mobile and web sessions.
- The calendar shall support reminders, upcoming visit cards, status indicators, and optional sync/export to approved external calendar channels where policy allows.
- Medical staff shall see appointment schedules relevant to their organization and role, while admins shall access booking analytics and operational schedule views.

9.4 Pharmacy & Medication Availability

- Patients shall search for a medication and see nearby pharmacies able to dispense it.
- System shall indicate availability status, estimated stock confidence, pharmacy hours, and insurance relevance where integrated.
- System shall display medication alternatives only where clinically approved and clearly labeled.

9.5 AI Chatbot / Agent

- Chatbot shall answer common questions about symptoms, specialties, hospital selection, appointments, insurance acceptance, medications, and platform usage.
- Chatbot shall retrieve answers from a curated knowledge base and approved medical guidance sources, not generate unrestricted clinical advice.
- Chatbot shall escalate urgent or red-flag symptom patterns to emergency guidance and recommend human care channels.
- Chatbot shall support Arabic and English.

9.5.1 Language and Localization

- All user portals shall support Arabic and English with a persistent language switch available from core navigation.
- The product shall store language preference at user-account and device level and remember it on subsequent sessions.
- Arabic views shall support proper right-to-left layout, localized terminology, and regionally appropriate formatting; English views shall use left-to-right layout.
- Chatbot and knowledge-base content shall be available in Arabic and English, with safe fallback when a translation is unavailable or not yet approved.

9.6 Medical Staff Workspace

- Medical staff shall search for a patient using National ID, Iqama, MRN alias, or demographics according to permission.
- Workspace shall show patient summary, timeline, active medications, allergies, recent encounters, and insurance details.
- Clinicians shall be able to add encounter notes, medication updates, care plans, and referrals where permitted.
- Pharmacists shall verify medication history and dispense-related status where enabled.

9.7 Admin Console

- Admins shall manage hospitals, clinics, pharmacies, doctors, staff accounts, and platform permissions.
- Admins shall review duplicate profile candidates, suspicious access patterns, and failed integrations.

- Admins shall configure knowledge base content for chatbot responses and escalation flows.
- Admins shall access analytics for adoption, search behavior, record linkage rate, and service quality.

10. User Journeys

10.1 Patient Journey

1. User signs up or logs in with National ID or Iqama, completes identity verification, selects language, and passes MFA.
2. User signs up or signs in with National ID or Iqama and identity verification.
3. User lands on unified health profile dashboard.
4. User searches symptoms or care need.
5. System recommends appropriate specialty, doctors, and nearby hospitals.
6. User books an appointment and sees it immediately on a synced calendar with reminders and status updates.
7. User checks pharmacy medication availability if a medication is prescribed or needed.
8. User uses chatbot for FAQs, navigation, and next-step guidance.

10.2 Medical Staff Journey

9. Staff signs in through organization-provisioned credentials, selects language if needed, and passes MFA before workspace access.
10. Staff authenticates with enterprise credentials and role-based access.
11. Staff searches patient by National ID or Iqama or approved identifiers.
12. Staff reviews longitudinal profile, prior medications, allergies, and history.
13. Staff documents encounter updates and care decisions.
14. Staff reviews relevant appointment schedule and upcoming calendar commitments for assigned clinics or departments.
15. System writes audit log for all sensitive record access and updates.

10.3 Admin Journey

16. Admin signs in through privileged account flow, completes MFA, and receives step-up authentication for high-risk operations.
17. Admin logs into admin console.
18. Admin monitors onboarding status of organizations and integrations.
19. Admin reviews data quality alerts, duplicate profiles, and access anomalies.
20. Admin updates provider listings, chatbot content, or permission policies.
21. Admin monitors booking performance, schedule utilization, and calendar-sync health.
22. Admin tracks KPIs, complaints, and platform operations.

11. Role-Based Access Model

Capability	Patient	Medical Staff	Admin
View own profile	Yes	No	Limited support view
View longitudinal patient record	No	Yes, permission-based	Yes, controlled and audited
Edit demographic data	Request correction only	Limited	Yes
Add clinical notes	No	Yes	No
Manage providers and hospitals	No	No	Yes
Moderate ratings or content	No	No	Yes

Access audit logs	No	Limited operational scope	Yes
Self registration / onboarding	Yes, identity-based	No, organization-provisioned	No, invitation-only
Login	Yes	Yes	Yes
MFA required	Yes	Yes	Yes, strongest policy
Book appointments	Yes	Limited operational view	Operational oversight
View synced calendar	Own appointments	Organization schedule scope	Operational calendar scope

- Patients shall use self-service registration tied to National ID or Iqama, verified contact channels, and MFA at sign in.
- Medical staff shall use organization-managed accounts with role assignment, staff-status verification, and MFA at each login.
- Admins shall use invitation-only privileged accounts with strongest MFA policy and step-up authentication for sensitive operations such as bulk export, permission changes, and provider approval.
- Access control shall combine role, organization, purpose, consent status, and contextual risk signals before exposing protected data.

12. Non-Functional Requirements

- High availability architecture with target uptime of 99.9 percent or better.
- Fast search performance for patient lookup and provider discovery.
- Strong encryption at rest and in transit.
- Comprehensive audit logging for every read or write action on sensitive health data.
- Granular role-based access control and least-privilege enforcement.
- Arabic-first and English-supported UX.
- Mobile-responsive web experience; native mobile apps may follow in later phases.
- Scalable architecture to support nationwide provider participation.
  - Role-specific sign up, login, recovery, and MFA flows must be available with high reliability and clear fallback procedures.
  - Authentication and authorization policy enforcement must support privileged-access management, step-up authentication, and suspicious-session termination.
  - Arabic / English language switching must be available on web surfaces with proper RTL / LTR rendering and terminology consistency.
  - Appointment booking and calendar synchronization must maintain consistency across confirmation, reschedule, cancellation, and reminder events.

13. Data Model (High-Level)

- Patient: identity, demographics, identifiers, dependents, insurance, contact data, consent flags.
- Clinical Record: diagnoses, medications, allergies, procedures, immunizations, encounters, attachments, provenance.
- Provider Organization: hospital, clinic, pharmacy, location, operating hours, insurance acceptance, ratings.
- Doctor: specialty profile, affiliations, credentials, languages, rating, schedule metadata.
- Medication Inventory Signal: medication code, pharmacy, stock status, timestamp, confidence.
- Chatbot Knowledge Object: FAQ, policy answer, clinical guidance snippet, escalation rule.
  - Hospital Master Registry: canonical facility record seeded from the uploaded Saudi hospital database with region, city, type, ownership, beds, website, notes, geocodes, and verification metadata.
  - Auth Identity Event: registration method, verified identifiers, login events, MFA method, device/session metadata, lockout state, and risk signals.

- Localization Preference: user language, chatbot language, content availability status, and translation approval metadata.
- Appointment: provider, patient, schedule slot, appointment type, status, reminder state, cancellation metadata, and linked calendar event identifiers.
- Calendar Event: appointment reference, start and end time, time zone, channel, notification status, sync state, and ownership scope.

14. Suggested System Architecture

Recommended stack based on your notes and enterprise fit:

Layer	Recommendation
Front End	React / Next.js web app for patients, medical staff portal, and admin console.
Back End	Node.js / NestJS or Java / Spring Boot microservices for identity, profile, provider search, medication availability, chatbot orchestration, and audit services.
Source Control	GitHub for repositories, CI/CD, environment workflows, and code review.
Documents & Object Storage	Secure document storage for attachments, referral letters, lab PDFs, imaging reports, and knowledge-base content.
Operational Database	Relational database such as PostgreSQL for core transactional data.
Primary Application Database Option	Supabase can be used as the rapid application backend for auth, storage, Postgres, and developer velocity in early phases.
Analytics Layer	Google Cloud analytics stack or equivalent for search analytics, operational dashboards, and usage telemetry, subject to data governance approval.
Search	Indexed search service for providers, doctors, hospitals, medications, and patient lookup metadata.
AI Layer	Retrieval-augmented chatbot service using approved FAQ content, product knowledge, and medically reviewed guidance.
Auth & Identity	Saudi identity-aware authentication service for sign up, login, MFA, session management, account recovery, device trust, and privileged-access controls.
Provider Master Registry	Hospital master directory seeded from the uploaded Saudi hospital database and normalized into a canonical registry with geocoding, deduplication, enrichment, and freshness tracking.
Scheduling & Calendar	Appointment booking engine, slot inventory, reminder service, cancellation / reschedule workflows, and synchronized calendar views for patients and provider teams.
Localization Layer	Arabic / English content management, language preference storage, translation workflow, and RTL / LTR presentation support across all portals.

15. Integration Requirements

- Hospital EMR or EHR integrations for demographics, encounters, medications, allergies, and summaries.
- Identity verification service for National ID and Iqama-based onboarding.

- Pharmacy systems for medication availability and dispense status.
- Insurance systems for eligibility and accepted-provider visibility.
- Maps and geolocation services for nearby hospital and pharmacy discovery.
- Notification service for SMS, email, and in-app updates.
- Analytics and monitoring tools for business and operational telemetry.
  - Hospital directory ingestion pipeline for batch import and normalization of the Saudi hospital registry seed dataset, with periodic verification and enrichment workflows.
  - Authentication providers for OTP, authenticator app, SMS or email verification, and enterprise identity federation where required for staff and admins.
  - Scheduling integrations for provider slot availability, appointment confirmations, cancellations, and reminders.
  - Calendar synchronization layer for in-app calendars and optional approved external calendar sync where policy allows.
  - Localization workflow for Arabic and English content approval across chatbot answers, product copy, and provider directory content.

16. Security, Privacy, and Compliance

- All sensitive health data must be encrypted at rest and in transit.
- Every data access event must be logged with user, time, action, and affected record.
- Patients should have clear visibility into consent, privacy notices, and where their data is used.
- Admin workflows must support account suspension, suspicious access review, and data correction operations.
- The platform should support data residency and regulatory requirements applicable to healthcare data in KSA.
- AI chatbot responses must be constrained, reviewable, and safe, especially for symptom-related guidance.
  - Multi-factor authentication must be mandatory for every role, with stronger enforcement for privileged users.
  - Patient sign up must include identity proofing against National ID or Iqama and verified contact-channel binding.
  - High-risk actions shall trigger step-up authentication and enhanced audit capture.

17. Ratings and Trust Model

- Hospitals and doctors may receive ratings and reviews subject to moderation and anti-abuse controls.
- Clinical quality metrics must not be inferred solely from consumer ratings.
- Verified visit linkage should be used where possible to increase trustworthiness of reviews.

18. MVP Scope

- Patient identity and unified profile foundation.
- Hospital and doctor discovery with ratings and filters.
- Basic pharmacy medication availability visibility.
- Medical staff read-centric longitudinal patient view with limited writeback.
- Admin portal for user, provider, and role management.
- Arabic and English FAQ chatbot trained on common questions and safe escalation logic.
  - Secure sign up, login, and MFA for patients, staff, and admins.
  - Saudi hospital directory seed ingestion and normalized hospital master registry.
  - Arabic / English language switch across patient, staff, admin, and chatbot journeys.

- Appointment booking and synced in-app calendar.

19. Post-MVP Enhancements

- Advanced referral routing, waitlist optimization, and cross-network rescheduling.
- Telemedicine visits and secure messaging.
- Lab and imaging result visualization.
- Predictive care reminders and chronic disease pathways.
- Claims and insurance preauthorization workflows.
- Family account management and dependent care coordination.
- Mobile apps for iOS and Android.

20. Success Metrics

- Duplicate patient profile reduction rate.
- Percentage of encounters with successfully linked longitudinal data.
- Doctor or hospital search-to-selection conversion rate.
- Average time for patients to find an appropriate provider.
- Medication search success rate and pharmacy match rate.
- Patient satisfaction and Net Promoter Score.
- Medical staff time saved during intake and record review.
- Chatbot containment rate versus human escalation rate.
- Security metrics: unauthorized access incidents, audit coverage, and policy violations.
  - Authentication completion rate, MFA success rate, and suspicious-login detection rate.
  - Hospital registry coverage, normalization accuracy, and percent of facilities verified or enriched beyond seed data.
  - Arabic / English experience parity and percent of interface verified for both language paths.
  - Appointment booking conversion rate, calendar sync success rate, and reminder delivery rate.

21. Risks and Mitigations

Risk	Impact	Mitigation
Data duplication and identity mismatch	Wrong or incomplete profile linkage	Master patient index, probabilistic matching, admin review queue
Hospital integration inconsistency	Incomplete records and poor trust	Standardized APIs, data contracts, staged onboarding, reconciliation jobs
Unsafe chatbot outputs	Clinical and reputational risk	Guardrails, curated knowledge base, escalation rules, human review
Low provider data quality	Poor discovery experience	Admin moderation, freshness checks, SLA reporting
Privacy or unauthorized access	Regulatory and trust impact	RBAC, audit logs, anomaly detection, strict operational controls
Weak authentication or account takeover	Unauthorized access to sensitive health data	Mandatory MFA, step-up authentication, anomaly detection, and account lockout.
Scheduling mismatch or calendar desync	Missed visits and poor patient trust	Transactional booking logic, sync monitoring, and reminder reconciliation.

22. Open Questions

- Will the platform operate as a national shared service, a private network product, or a consortium-based exchange?
- What exact data-sharing consent model will govern cross-hospital visibility?
- What interoperability standard and clinical coding systems will be mandatory for participating organizations?
- Will medication availability be real-time inventory, delayed sync, or confidence-based estimates?
- Will doctor matching by symptoms remain rules-based in MVP or include machine learning from launch?
- Which entities can write back to the unified longitudinal record, and under what governance?

23. Recommended Delivery Roadmap

23. Phase 0: Product definition, legal and compliance alignment, identity strategy, architecture design.
24. Phase 1: Unified identity, patient profile, provider directory, admin console foundation.
25. Phase 2: Medical staff workspace, hospital integrations, medication availability integrations.
26. Phase 3: AI chatbot or agent, symptom-to-specialty matching, analytics dashboards.
27. Phase 4: Appointments, telemedicine, deeper insurer integrations, advanced care coordination.

Appendix A. Saudi Hospital Directory Seed Dataset

The uploaded Saudi hospital file should be used as the initial national hospital registry seed for Wiaam. The source exposes core directory attributes such as facility name, location, region, type, ownership, bed count, website, and notes, which should be normalized before production use.

Source file fields from the uploaded directory should map as follows:

Source Field	Platform Field	Usage in Wiaam
اسم المنشأة	facility_name_ar	Canonical Arabic hospital or facility name
الموقع	city_or_locality	City / locality used in discovery and search filters
المنطقة	region	Region-based filtering, analytics, and coverage views
النوع	facility_type	General, specialist, university, military, etc.
الملكية	ownership_type	Government, private, military, university, or mixed ownership
عدد الأسرة	licensed_beds	Capacity indicator where present
الموقع الإلكتروني	website	Directory linkout and verification support
ملاحظات	source_notes	Operational notes and curation context

24. Final Product Statement

Wiaam should be positioned as the healthcare orchestration layer that sits above fragmented provider systems and turns disconnected records into a secure, navigable, patient-centered experience for Saudi Arabia. Its differentiator is not only unified records, but unified action: find care, verify providers, locate medicine, understand next steps, and give every authorized stakeholder the right view at the right moment.